TENDER FOR SUPPLY, INSTALLATION AND COMMISSIONING OF INDUSTRIAL ELECTRICAL CONTROL PANEL

Tender Reference No: SISAP/DEMO/CP-2026/01

Issuing Authority: Procurement Division, State Public Works Department (demonstration document)

1. SCOPE OF SUPPLY

1.1 The scope of this tender covers design, manufacture, supply, installation, testing and commissioning of a floor mounted industrial electrical control panel for a 415 V, 50 Hz, three phase power distribution and motor control application.

1.2 The control panel shall be suitable for indoor industrial installation and shall house incoming switchgear, outgoing feeders, busbars, protective devices and a control section.

1.3 The purchaser reserves the right to inspect the panel at the manufacturer's works before dispatch.

2. TECHNICAL SPECIFICATION

2.1 The control panel shall be a low voltage switchgear and controlgear assembly rated for a system voltage of 415 V, 50 Hz and a busbar current rating of not less than 800 A.

2.2 The panel shall be designed for a short circuit withstand strength of 50 kA for one second.

2.3 The enclosure shall be fabricated from sheet steel of not less than 2 mm thickness and shall be provided with a degree of protection of IP54 for indoor use. The enclosure finish shall be powder coated after phosphate pretreatment.

2.4 Busbars shall be of aluminium or copper and shall be adequately supported to withstand the short circuit forces. Creepage distances and clearances shall be maintained in accordance with the applicable standard for the degree of pollution expected.

2.5 Temperature rise limits of all current carrying parts shall be within the limits specified for the applicable class of equipment.

2.6 The panel shall be provided with an earthing busbar of adequate cross section, and all metallic parts of the enclosure shall be bonded to the earthing system.

2.7 The ambient conditions at site are: temperature range 5 degrees C to 45 degrees C, relative humidity up to 95 percent, and altitude below 1000 m above mean sea level.

3. PROTECTIVE DEVICES

3.1 The incoming feeder shall be provided with a moulded case circuit breaker or a switch fuse unit of suitable breaking capacity.

3.2 Outgoing feeders shall be protected by miniature circuit breakers with suitable current ratings and tripping characteristics.

3.3 All protective devices shall have a rated short circuit capacity adequate for the fault level at the point of installation.

4. SAFETY REQUIREMENTS

4.1 The control panel shall provide protection against direct contact and indirect contact in accordance with the applicable code of practice for electrical installations.

4.2 All doors and covers shall be interlocked or provided with means of preventing inadvertent contact with live parts.

4.3 Earthing shall be provided in accordance with the applicable code of practice for earthing, and earth continuity shall be verified before energisation.

4.4 The panel shall be installed with adequate ventilation clearance and shall not obstruct access for maintenance.

5. TESTING AND INSPECTION

5.1 All routine tests shall be carried out at the manufacturer's works and test certificates shall be furnished.

5.2 Type test reports for the assembly shall be made available for scrutiny on request.

5.3 A high voltage test and an insulation resistance test shall be performed before the panel is energised at site.

6. INSTALLATION AND COMMISSIONING

6.1 The panel shall be installed on a foundation suitable for the equipment, with cable entry from the bottom unless otherwise agreed.

6.2 Cable glands, lugs and terminations shall be provided by the supplier.

6.3 The installation shall be carried out in accordance with the applicable code of practice for electrical wiring installations.

7. DRAWINGS AND DOCUMENTATION

7.1 The supplier shall furnish dimensional drawings, schematic diagrams and a general arrangement drawing showing clearances for approval before manufacture.

7.2 As built drawings and an operation and maintenance manual shall be supplied on completion.

8. WARRANTY

8.1 The control panel shall be warranted against defects in material and workmanship for a period of 24 months from the date of commissioning.

Note for evaluators: this is a demonstration tender prepared for SISAP. It intentionally omits certain clauses (for example a stated certification requirement) so that the specification gap detector has something to report.